

SMARTBRIDGE SKILL WALLET

Scalability & Future Plan

PocketSmart AI — Page 1

Date: 29 September 2026	Team ID: SWTID-2026-5955
Project Name: PocketSmart AI — Your Smart Budget & Recommendation Assistant	Maximum Marks: 1 Mark



Overview

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

1. Current System Limitations

S.No	Limitation	Priority to Address (High/Medium/Low)	
1	Single Gemini API key rate limit 60 req/min	Throttles under 100+ concurrent users, response delay 3–5 sec	High
2	SQLite file-based storage not scalable	Performance degradation beyond 10k users, no horizontal scaling	High
3	No caching layer for repeated budget queries	Same budget "50000 Home" query hits Gemini again — waste cost & latency	Medium

2. Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask instance on Render free tier — 50–100 users	Horizontal scaling with Docker + Kubernetes; Load balancer (Nginx); Auto-scaling on IBM Cloud Code Engine
2	Data Storage	SQLite local file	Migrate to PostgreSQL on IBM Cloud Databases + Redis for session cache + S3 for jewelry image uploads
3	Performance	Avg 1.4 sec, max 3.8 sec, no caching	Implement OAuth2 (Google); Rate limiting via Flask-Limiter; AES-256 encryption for history; Audit logs Pillow resize to 512×512
4	Security	Basic JWT + b2chpt	

Future Roadmap:

Phase	Planned Feature/Enhancement	Target Timeline	Expected Impact
Phase 1 (Immediate - 1 month)	• Real e-commerce API integration: Amazon Product API, Flipkart Affiliate API, IKEA API for live price fetching • Enhanced Gemini prompt caching • Budget history export: CSV, PDF	Q4 2026	• Live pricing accuracy • Reduce mock data • Cost saving: 40%
Phase 2 (Short-term - 3 months)	• New category: Travel Planner budget • Furniture AR visualization • Voice assistant for budget query • Flutter mobile app • Multi-language support: Hindi, Tamil, Telugu	Q1 2027	• Expand user base to 50k • Accessibility for rural users • Mobile reach
Phase 3 (Mid-term - 6 months)	• AI-powered negotiation tips • Price drop alerts via email & WhatsApp • Collaborative family budgeting: shared wallets • Payment gateway: Razorpay for direct purchase • ML model for spending pattern analysis	Q2 2027	• Monetization via affiliate commission: 5-10% • Family retention: +35%
Phase 4 (Long-term - 12 months)	• Enterprise version for interior designers & party planners • Blockchain for transparent budget tracking • Partnership with IKEA, Swiggy, OYO for exclusive discounts • Global expansion: USD & EUR currency support • Advanced analytics dashboard with charts	Q4 2027	• B2B revenue stream • Brand partnerships • Scale to 500k users • IPO readiness